FLAIR: A Federated Benchmark for ICU Clinical Decision Support Across Multiple Academic Medical Centers Using the Common Longitudinal ICU Format

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Abstract

Background. AI clinical decision support (CDSS) tools are proliferating in practice, yet external validation remains concentrated on a handful of public ICU electronic health record datasets. Three problems result: (i) *benchmark saturation*, repeated tuning on the same public test sets inflates reported performance; (ii) a *reproducibility gap*, few publications release runnable code, so retrospective re-execution at a new site is slow or infeasible; and (iii) *incomplete reporting*, most studies omit full TRIPOD+AI items and subgroup fairness metrics that regulators increasingly require. Bilateral data use agreements stretch multi-site validation over years and rarely reach more than a few sites.

Methods. FLAIR is a federated benchmark in which code travels to data, never the reverse. It is anchored on the Common Longitudinal ICU Format (CLIF), a common data element–aligned critical-care schema adopted by 12+ academic medical centers. Five binary classification tasks span outcome, deterioration, and intervention targets (ICU daily mortality, daily discharge to long-term acute care, extubation failure, sepsis, unplanned ICU readmission). Models run as committee-reviewed code or containers at each site and emit aggregate-only JSON reports (cohort characteristics, discrimination, calibration, and subgroup fairness metrics following TRIPOD+AI) with HIPAA cell suppression. The full FLAIR stack, library, task definitions, container/code specification, and report generators, is released open source (Apache 2.0) at `github.com/clif-consortium/FLAIR`, and the task catalog is designed to grow: any contributor can add a new benchmark task as a single-file Python module via GitHub pull request.

Results. Mode-matched baselines trained on MIMIC-CLIF discriminate well on internal test (landmark mortality AUROC 0.86, sepsis peak 0.69). At the first external site (Rush, 13,839 ICU stays) discrimination drops on every task, from a modest decline on mortality (0.86→0.75) to a collapse on sepsis (0.69→0.41), quantifying the cross-site distribution shift the benchmark exists to surface.

Conclusions. FLAIR shrinks retrospective multi-site CDSS validation from years to weeks, exposes models to a growing federated cohort that resists public-benchmark saturation, and produces the TRIPOD+AI and fairness evidence package regulators increasingly require, without patient-level data leaving any site. The initial five-task suite is a seed, not a ceiling: the open governance model invites the community to extend the benchmark to new clinical questions over time.

1 Introduction

The bottleneck in clinical decision support (CDS) for the intensive care unit is no longer the supply of models. Predictive algorithms for sepsis, acute kidney injury, extubation readiness, and ICU mortality have been published at a rate the literature struggles to absorb. What the field is missing is the evidence that any given model behaves acceptably outside the institution that built it. The body of evidence required to support routine clinical use has not kept pace with the proliferation of models, and that gap is, in our view, primarily an *access* gap: representative, multi-institutional data on which to test a model rarely sit within reach of the team that needs them.

The evidence gap is an access gap. External validation of even flagship EHR-based CDS has remained stubbornly narrow. The Epic Sepsis Model, deployed in hundreds of United States hospitals, was first independently evaluated at exactly one site, with discrimination falling to AUC 0.63 against vendor-claimed ranges of 0.76–0.83 (Wong et al., 2021; Habib et al., 2021). Duke's Sepsis Watch, an exemplar academic deployment, took more than six years to be assessed at its first external site (Sendak et al., 2025). The Targeted Real-time Early Warning System (TREWS) reached five sites, all within a single health system (Adams et al., 2022). Mayo Clinic's electrocardiogram-based low-ejection-fraction screen, taken through the FDA pathway, required multi-year coordination to cover roughly four to six external sites (Carter et al., 2025). Each of these efforts required a bilateral data use agreement, shipping de-identified patient data to a coordinating analytic center. None scale to dozens of sites at the pace the rest of the AI ecosystem moves.

Public benchmarks saturate; private benchmarks don't scale. The community has filled some of the gap with open ICU benchmarks (van de Water et al., 2023; Yèche et al., 2021) built on MIMIC, eICU, HiRID, and AUMCdb. These tools are essential for development but inherit the selection biases of the source datasets (Röösli et al., 2022), and the literature now documents how repeated tuning against a fixed public benchmark produces "illusory generalizability" (Recht et al., 2019; Chekroud et al., 2024). Federation-by-data-use- agreement helps but is narrow in scope: the Observational Health Data Sciences and Informatics network federates SQL queries and coefficient sharing for logistic models on a pre-harmonized common data model (Reps et al., 2020); MedPerf and the FeTS Challenge prove that a portal-orchestrated, model-to-data benchmarking ecosystem works at scale, but for imaging (Karargyris et al., 2023; Pati et al., 2022). None of these combine modern, longitudinal EHR-based CDS, a clinically meaningful ICU schema, and an extensible task catalog open to the community.

Contribution. We present FLAIR (Federated Learning Assessment for ICU Research), a benchmark framework with three load-bearing properties. First, it operates on the *model-to-data* paradigm pioneered by the DREAM Challenges and applied at scale to EHRs in the EHR DREAM Challenge (Guinney and Saez-Rodriguez, 2018; Bergquist et al., 2024), but extended from one-time competitions to a standing, repeatable infrastructure. Second, it is anchored on the Common Longitudinal ICU Format (CLIF), a critical-care common data element schema adopted by 12+ academic medical centers (Rojas et al., 2025), conceptually parallel to mCODE for oncology (HL7 International, 2024) and complementary to the Medical Event Data Standard (Arnrich et al., 2025). Third, FLAIR is released open source (Apache 2.0, `github.com/clif-consortium/FLAIR`): the task catalog, evaluation protocol, and submission contract are extensible by GitHub pull request, with each new task added as a single Python module under `flair_benchmark/tasks/`. The benchmark

co-evolves with clinical need rather than ossifying around a single author group's priors, and the five-task v1 suite below is explicitly a seed for that catalog rather than a fixed scope.

We organize the rest of the manuscript around three commitments: the CLIF-anchored *datasets* that participating sites contribute (§3), the initial five-task *benchmark* suite together with the open contribution protocol for adding more tasks (§4), and the federated submission and evaluation *protocol* (§5).

2 Methods

We describe FLAIR as three coupled artifacts: a data layer (CLIF, with MIMIC-CLIF as the public development entry point), a task layer (five benchmark definitions implemented as single-file Python modules), and a protocol layer (model-to-data execution as code or container, with aggregate-only JSON reporting). Detailed specification is deferred to §3–§5; per-task algorithmic schematics are in the Supplementary Appendix.

Reporting follows the TRIPOD+AI statement (Collins et al., 2024); the FLAIR library populates the applicable checklist items through its per-task report bundle of discrimination, calibration, decision-curve, and subgroup-fairness JSON files. A Datasheet for Datasets (Gebru et al., 2021) accompanies the public MIMIC-CLIF cohorts, and a Model Card (Mitchell et al., 2019) accompanies the baseline gradient-boosted-tree models.

Institutional review. Each participating consortium site obtained independent institutional review board (IRB) approval for participation in CLIF and for execution of FLAIR benchmark code on its patient data. No patient-level data are transmitted between sites; only aggregate JSON outputs leave each institution. Site governance retains final approval over which submitted models are executed.

Data availability. MIMIC-CLIF is publicly accessible via PhysioNet under the standard MIMIC data use agreement. Consortium data are not redistributable; they are accessible only by submitting code through the FLAIR portal for federated execution at participating sites.

Code availability. The FLAIR library, task definitions, report generators, and the CLIF-WorkBench container specification are released under Apache 2.0 at `github.com/clif-consortium/FLAIR`.

3 Datasets

The Common Longitudinal ICU Format

CLIF is a critical-care minimum dataset that harmonizes ICU electronic health record content into a small set of relational tables with standardized vocabularies for vital sign and laboratory categories, medication routes, and respiratory support modes (Rojas et al., 2025). As of the founding paper, CLIF spanned ∼9 health systems, ∼39 hospitals, and ∼111,000 ICU admissions; the consortium has since grown to 12+ academic medical centers covering ∼2,000 ICU beds. The

schema is intentionally narrow, vitals, labs, medications, respiratory support, admission–discharge–transfer, and patient demographics, so that contributing sites can map their EHR exports once and reuse the mapping across studies. In spirit, CLIF is to critical care what mCODE (HL7 International, 2024) is to oncology: a clinically scoped common-data-element layer that makes downstream analytic code portable. CLIF is complementary to broader EHR standards such as the Medical Event Data Standard (MEDS) (Arnrich et al., 2025) and OMOP. FLAIR lets a developer consume CLIF data in either of two equivalent ways, chosen per project: natively, as CLIF relational tables through the `clifpy` interface, or as a MEDS event stream via the maintained CLIF→MEDS bridge (`CLIF_MEDS`, `github.com/Common-Longitudinal-ICU-data-Format/CLIF_MEDS`), so adoption never requires schema lock-in. Crucially, because CLIF standardizes the data *elements* themselves through its minimum common ICU data elements (mCIDE), not merely the table layout, transportability is not only syntactic: a model meets the same coded vocabularies at every site, so FLAIR measures genuine cross-site generalizability of trained models rather than an artifact of differing feature definitions.

MIMIC-CLIF as the public development entry point

To support open development, the consortium distributes MIMIC-CLIF: the MIMIC-IV critical-care dataset (Johnson et al., 2023, 2016) re-expressed in the CLIF v2.1 schema, with ∼70,000 ICU stays. Method developers train and self-evaluate against MIMIC-CLIF; because the schema is identical to consortium-side data, the same code runs unmodified when later executed federated. This mirrors the "develop-on-public, validate-on-private" pattern that has become canonical in clinical AI but is rarely operationalized end-to-end.

Consortium cohort

Coverage. Across the 12+ sites currently in CLIF, FLAIR operates on an aggregate cohort of ∼500,000 ICU stays spanning urban academic medical centers, large community hospitals, and health systems with mixed payer populations. Demographic distribution (sex, race, ethnicity, age), full-hospitalization SOFA (worst-in-window), and Charlson and Elixhauser comorbidity indices are summarized per task in the cohort report (`table1.json`). Table 2 reports the MIMIC-CLIF entry-point cohort per task. Table 1 gives the federated consortium skeleton: MIMIC-CLIF is populated as the public development entry point, and each participating site contributes the same suppressed `table1.json` from its own data; site columns read `TBD` until the federated run completes.

	MIMIC-CLIF	**Rush**	**Site C**	**Site D**
ICU stays	72,778	13,839	TBD	TBD
Female (%)	43.2%	47.5%	TBD	TBD
Age, median (yr)	67	65	TBD	TBD
SOFA hosp. max (med.)	5	4	TBD	TBD

Table 1: Federated consortium cohort skeleton (one suppressed `table1.json` per site, T1 cohort). MIMIC-CLIF is the public development entry point; Rush is the first reporting external site. Remaining sites report at submission, so their cells read `TBD`. Cell suppression at $n<10$ is applied site by site.

	T1 mort.	T2 LTACH	T3 ext-fail	T4 sepsis	T5 readmit
ICU encounters	64,408	64,408	26,768	80,600	67,235
Decision rows	198,125	198,125	26,771	14,347,321	67,250
Positive rate (row)	19.2%	14.3%	5.4%	0.2%	9.2%
Stays ever positive	13.0%	6.3%	5.4%	7.2%	9.2%
Sex: F / M	27,795/36,613	27,795/36,613	10,031/16,737	35,760/44,840	29,199/38,036
Age 65+ share	55.3%	55.3%	54.1%	54.8%	54.1%
SOFA hosp. max (med., IQR)	5 (2–8)	5 (2–8)	7 (5–10)	4 (2–7)	4 (2–7)
Charlson (median, IQR)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)

Table 2: MIMIC-CLIF cohort snapshot per task, written as a suppressed `table1.json` by the FLAIR baseline (`flair-baseline train`; `flair_benchmark` v0.1.0; HIPAA cell-suppression at $n<10$). *Decision rows* are one-per-stay for T3 (extubation episode) and T5 (ICU readmission, anchored at first-ICU discharge), and one-per-window otherwise (daily at 07:00 local for T1/T2; hourly for T4). Counts are per stitched encounter (`hospitalization_join_id`; hospitalizations $\leq$6 h apart are merged), so *ICU encounters* $\leq$ the raw hospitalization count. *Stays ever positive* equals the row-level rate for T3 and T5 because those tasks are one-per-stay. *SOFA hosp. max* is the worst SOFA score across the full hospitalization [*admission_dttm*, *discharge_dttm*], computed via `clifpy`'s polars backend (`compute_sofa_polars`). Charlson and Elixhauser are computed from `HospitalDiagnosis`; 49–54% of stays are missing comorbidity scores in the MIMIC-CLIF export, a known data-quality gap (Limitations).

Privacy-preserving access. Three properties together constitute FLAIR's access model. *No patient-level data leaves any site*: outputs are JSON aggregates produced by the FLAIR library inside the container. *Library-enforced suppression*: counts below 10 in cohort tables become `"<10"` in `table1.json`; demographic subgroups with $n < 10$ are dropped entirely from `fairness.json`; reliability bins with fewer than 10 observations report a null event rate in `calibration.json`. *Pre-release governance review*: before any submission is dispatched to sites, the Benchmark Committee security-reviews the submitted code and model and validates the task code against synthetic (fake) data, so only audited artifacts, whether shipped as raw code or wrapped in a container, ever execute against patient data, consistent with the spirit of analysis platforms such as OpenSAFELY (Collaborative, 2021) in which the code travels and the data does not. Together these properties move evidence generation from the slow, bilateral DUA path to a routine, repeatable operation.

4 Benchmarks

Task taxonomy

Every FLAIR task is declared on three axes in a single `META` dictionary: *task class* (*outcome*, *deterioration*, *intervention*, or *operational*); *temporal pattern* (*episode*, *continuous*, or *fixed horizon*); and *prediction unit* (*one-per-stay* or *one-per-window*). The triple determines the shape of the cohort and prediction parquets, which fields are mandatory, and which evaluation primitives the report builder invokes. New task authors fill the triple before writing any SQL or feature code; reviewers consume it before opening a single line of the implementation. The taxonomy is deliberately small so that the boundary between "new task" and "existing task" is unambiguous.

Initial five-task suite

The v1 release covers three outcome targets and two intervention-anchored targets spanning ICU operations, disposition, and infectious physiology.

- **T1, ICU Daily In-Hospital Mortality (07:00 local).** One-per-window, 24-hour input window anchored at daily 07:00. Cohort: first ICU stay with discharge in {long-term acute care, mortality}. Label: in-hospital death or transfer to hospice during that ICU stay.
- **T2, ICU Daily Long-Term Acute Care Discharge (07:00 local).** Complementary label to T1 on the same daily grid: prediction that the patient will be discharged to long-term acute care rather than dying or being discharged home. Together T1 and T2 capture the disposition decision that anchors much downstream ICU planning.
- **T3, Extubation Failure within 24 h.** One-per-stay, anchored at the first confirmed invasive-mechanical-ventilation extubation event identified by the consensus-window episode detector. Label: reintubation within 24 hours; death without reintubation in that window is excluded to avoid palliative extubations. Benchmark targets the classic respiratory liberation question (Thille et al., 2011).
- **T4, Sepsis (antibiotic-anchored) within 6 h.** One-per-window on an hourly grid; t_{zero} is the first qualifying systemic intravenous antibiotic. Cohort follows the Sepsis-3 hospital-wide pattern (Singer et al., 2016; Seymour et al., 2016; Rhee et al., 2019); leakage prevention drops decision points at or after t_{zero}. The implementation mirrors the antibiotic-anchored evaluation framing of Kamran et al. (2024).
- **T5, Unplanned ICU Readmission.** One-per-stay, anchored at first-ICU discharge; the label is a return to the ICU later in the same hospitalization through a non-procedural path. Working from the CLIF `ADT` location stream, planned ICU → procedural → ICU excursions are merged and excluded; the cohort drops death-in-ICU, direct ICU discharge, sub-hour first stays, and intermediate ER visits. The framing follows structured-EHR ICU-readmission prediction (Rojas et al., 2018).

Each cohort parquet carries a fixed ten-column schema, identifiers, window endpoints, label, split, age, sex, race, and ethnicity, so that downstream feature engineering and evaluation code is shared across tasks.

Count-based reference baselines

The leaderboard is anchored by a count-featurization baseline in the EHRSHOT/FEMR tradition (Wornow et al., 2023; Steinberg et al., 2021), adapted to CLIF. Every row of a CLIF event table (`vitals`, `labs`, `medication_admin_*`, `respiratory_support`) is mapped to a deterministic `<table>.<category>` token (e.g., `vitals.heart_rate`, `labs.creatinine`), and each cohort window is reduced to a sparse integer count vector over those tokens. The featurizer is config-free and exhaustive, every schema-locked category flows through with no per-task curation, so the same code runs unchanged on MIMIC-CLIF and at every consortium site. A single XGBoost head consumes the count vectors and emits a probability score `y_prob`. It is trained on MIMIC-CLIF and shipped frozen to sites; hyperparameter grids, the token dictionary, and the missingness convention (absent token = zero count) are given in the supplementary appendix (Sec. A). Internal-test performance on

Table 3: Federated leaderboard skeleton: XGBoost count-featurization baseline, headline *mode-matched* AUROC (the deployed, stay-level metric, §4) with 1,000-iteration bootstrap 95% CI. MIMIC-CLIF is the internal test split (75/25 hold-out on stitched encounters, `hospitalization_join_id`, so an encounter never spans train and test); Rush is the first external site, the MIMIC-trained model scored on a deterministic 25% hold-out of Rush data; remaining site columns read `TBD`. Full internal detail, including row-level AUROC and AUPRC, is in Table 5.

Task	Mode	MIMIC-CLIF AUROC (95% CI)	Rush AUROC (95% CI)	Site C	Site D
T1, ICU daily mortality	landmark	0.864 (0.856–0.872)	0.748 (0.735–0.763)	TBD	TBD
T2, LTAC discharge	landmark	0.789 (0.775–0.803)	0.658 (0.622–0.691)	TBD	TBD
T3, Extubation failure 24 h	episodic	0.696 (0.665–0.719)	0.565 (0.522–0.605)	TBD	TBD
T4, Sepsis 6 h	peak	0.693 (0.678–0.709)	0.407 (0.391–0.421)	TBD	TBD
T5, Unplanned ICU readmission	episodic	0.641 (0.626–0.658)	0.509 (0.489–0.531)	TBD	TBD

MIMIC-CLIF is summarized in Table 3; the per-task clinical thresholds, alarm directions, and report modes are single-sourced in one registry (`flair_benchmark/_constants.py`) and folded into each task's audit hash, so a policy change is visible at every site. At the first external site (Rush), the same MIMIC-trained models lose discrimination across every task, from a modest drop on landmark mortality (0.864 → 0.748) to a collapse on sepsis (peak AUROC 0.693 → 0.407); this cross-site degradation is the failure mode FLAIR exists to surface.

Evaluation modes

A task's temporal pattern dictates how its predictions are scored, so the report builder runs in one of three *modes*, single-sourced per task in the policy registry. **Episodic** (T3, T5, one-per-stay) scores one row per stay directly. **Peak** (T4, hourly screening) collapses a stay's many windows to its single highest risk before scoring, the unit of action for a screening alarm; per-window pooling would inflate apparent skill, and naive recalibration on the peak is ill-posed, so calibration and decision curves are omitted (the "peak-trap"). **Landmark** (T1, T2, daily) scores a sequence of lead-time risk sets and reports a lead-time profile, the early-warning view. The distinction is load-bearing: the sepsis baseline scores row-level AUROC 0.82 but mode-matched peak AUROC 0.69, and the mode-matched number is the one a unit would experience. Mode-matched AUROC is therefore the headline metric in Table 3.

Leaderboard mechanics

For each task and site, FLAIR reports area under the receiver-operating characteristic curve (AUROC), area under the precision–recall curve (AUPRC), and F_1 at the operating threshold, each with 1,000-iteration bootstrap 95% confidence intervals, in `discrimination.json`. Calibration is summarized by slope, intercept, and Brier score, with raw reliability points stored in `calibration.json` so that the central dashboard can render reliability and ROC/PR curves without re-touching patient data. Fairness is reported as per-subgroup AUROC across sex, race, ethnicity, and age bins together with explicit parity gaps. The leaderboard ranks submissions on the primary metric for each task and displays per-site variance prominently: a single high pooled AUROC hides exactly the failure mode FLAIR exists to surface.

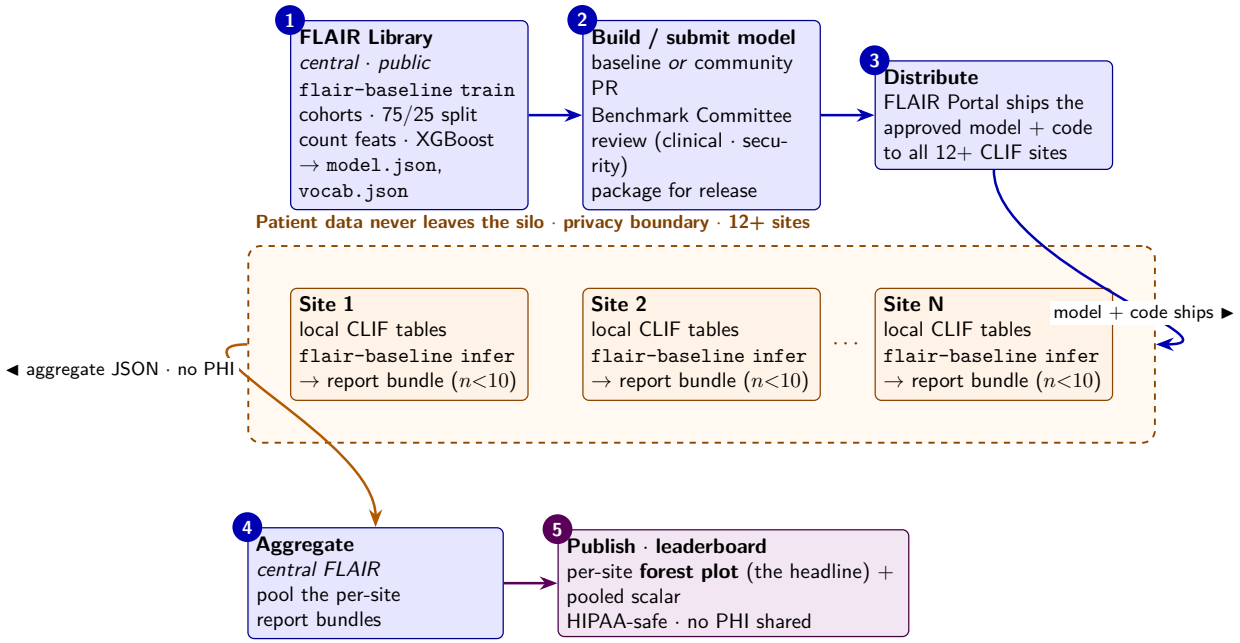


Figure 1: **The FLAIR model-to-data loop.** Patient-level data never leaves any site; only a model crosses in and an aggregate report crosses out. *Develop (central, public):* a model is built against MIMIC-CLIF with `flair-baseline train` (cohorts, a 75/25 split on `hospitalization_join_id`, count features, an XGBoost head) or submitted by the community as a pull request; the Benchmark Committee reviews it for clinical and security soundness. *Ship:* the FLAIR Portal distributes the approved model folder and code to the 12+ CLIF sites. *Run locally:* each Site Lead / Data Steward applies it with `flair-baseline infer` on the site's own CLIF tables, which writes three site-prefixed folders, and returns only the non-PHI **report bundle**, a *mode-keyed* set of JSONs (`discrimination`, `calibration`, `dca`, `fairness`, and, for landmark tasks, `leadtime`) plus `table1.json` and a suppressed code registry, with cell suppression at $n < 10$ — metrics only, no patient rows. *Aggregate & publish:* the Portal pools the per-site reports into the public leaderboard, where the per-site **forest plot**, not the pooled scalar, is the headline signal.

An open, extensible task catalog

FLAIR is released open source under the Apache 2.0 license at `github.com/clif-consortium/FLAIR`, and the five tasks above are explicitly the *seed* of a growing catalog rather than the benchmark's permanent scope. Any researcher or clinician can propose a new task as a single Python module under `flair_benchmark/tasks/` containing the `META` dictionary and a `build` function returning the canonical cohort frame; no registry edits or library changes are required. Acceptance review verifies (i) reproducibility against MIMIC-CLIF, (ii) clinical face validity through a clinician-led review, and (iii) site-governance acceptance at a minimum of two consortium sites before the task enters the federated suite. This is the structural counter to benchmark saturation: the catalog grows with community need, distributed across many proposers, rather than ossifying around the priors of any single author group (Recht et al., 2019; Chekroud et al., 2024).

5 Protocols

The federated protocol is what makes FLAIR's privacy properties operational. We first name the actors that operate the ecosystem, then describe the protocol in three steps, developer-side submission, site-side execution, and central aggregation, and position it against the closest precedents.

The FLAIR ecosystem: roles and the Portal

FLAIR is operated as an ecosystem of four roles, mirroring the model-to-data benchmarking pattern MedPerf established for medical imaging (Karargyris et al., 2023) and bringing it to critical-care longitudinal EHR. The *FLAIR Portal* is the orchestration layer: it registers tasks, models, and sites, distributes the containerised code and Dockerfiles, collects the returned aggregate JSONs, and renders the leaderboard. A *Benchmark Committee*, clinicians, methodologists, regulators, and patient representatives, reviews each new task, which enters as a single-file Python module via GitHub pull request, for reproducibility against MIMIC-CLIF and clinical face validity. The same Committee is the privacy and safety guarantee on the model side: before a submission is released to sites it security-reviews the submitted code and model and validates the task code against synthetic (fake) data, so only audited artifacts are ever dispatched, whether shipped as raw code or wrapped in a container. A *Model Owner* submits a frozen model, as code or wrapped in a container. At each institution a *Site Lead / Data Steward* pulls the approved submission from the Portal, runs it in CLIF-WorkBench against local CLIF tables, and returns only the standardized non-PHI report bundle. The lifecycle is therefore register → approve → execute → release: no patient-level data leaves any site, and every site runs byte-identical Portal-distributed code.

Submission

A clinical decision support developer using FLAIR works entirely against MIMIC-CLIF during development. The library exposes CLIF tables through `clifpy`, the canonical CLIF Python interface. The benchmark CLI exposes `flair load-task` (build a task-specific cohort parquet) and `flair build-report` (score predictions into the report bundle); the reference baseline wraps these as `flair-baseline train` and `flair-baseline infer`. Each run writes three site-prefixed folders, separating PHI (cohort, events, predictions; stays local) from the non-PHI upload bundle and the model folder that ships to other sites. Submission to the consortium consists of code and a frozen model artifact, optionally packaged as a container image (Docker or Singularity) that wraps both. The model-to-data paradigm itself is not new, it was introduced by the DREAM Challenges (Guinney and Saez-Rodriguez, 2018) and demonstrated for EHR-based mortality prediction at scale in the EHR DREAM Challenge (Bergquist et al., 2024). FLAIR's contribution is to move it from a one-time competition format into a standing benchmark and submission service. Before any model is dispatched to a site, the Portal runs its container against MIMIC-CLIF as a compatibility check, so only models that execute cleanly on the schema-locked public data are approved for federated execution.

Execution

Each consortium site runs the committee-reviewed, security-approved submission, raw code or a container, inside the CLIF-WorkBench environment. Because every submission has already passed central security review of its code and model and a synthetic-data validation pass before release,

Table 4: FLAIR alongside adjacent federated infrastructures.

	OHDSI/LEGEND	MedPerf	FeTS	OpenSAFELY	**FLAIR**
Data model	OMOP	task-specific	imaging (brain MRI)	TPP / EMIS	CLIF
Federation unit	SQL / coefficients	container	training gradients	analytic code	container / model code
Modern DL CDS	no	yes (imaging)	yes (training only)	limited	yes (longitudinal EHR)
Hosted-model portal	no	yes	yes	no	yes
PR-extensible tasks	no	partial	no	n/a	yes
Sites (current)	many	∼10s	71	English NHS	12+

sites execute only audited artifacts. The submitted code is permitted to read the site's CLIF tables, write a predictions parquet (cohort plus `y_true`, `y_pred`, `y_prob`), and invoke the FLAIR report builder, which produces a mode-keyed report bundle (`discrimination`, `calibration`, `dca`, `fairness`, and `leadtime` for landmark tasks; peak tasks omit calibration and decision curves). Every JSON carries the FLAIR library version, the generation timestamp, and the task audit hash. Sites retain final approval over which models execute; the simplicity of the per-task module, one Python file expressing cohort, label, and exclusions in a few hundred lines, makes that pre-execution audit tractable.

Aggregation, leaderboard, and lifecycle

The central aggregator merges the per-site aggregate JSONs for a given task version and publishes per-site and pooled metrics. Per-site variance is the primary signal of generalization; pooled metrics are reported but de-emphasized. The leaderboard records both retrospective benchmark performance and, for tasks where sites have opted in, prospective silent-trial evaluation under emerging guidance for clinical AI preregistration (Goodman et al., 2025; Joshi et al., 2025). Recent work on operational templates for monitoring AI in routine clinical practice (NEJM AI Pragmatic Trial Authors, 2024) informs the prospective arm.

Comparison with adjacent federated infrastructures. The closest precedent for federated multi-site evaluation in healthcare is the OHDSI network, whose LEGEND framework standardized network studies on the OMOP common data model (Reps et al., 2020); LEGEND federates SQL queries and coefficient sharing for logistic models. MedPerf (Karargyris et al., 2023) demonstrates federated benchmarking for medical imaging models packaged as containers; the FeTS Challenge (Pati et al., 2022) established federated training, rather than validation, as a viable mode for imaging. OpenSAFELY (Collaborative, 2021) delivers a code-travels-not-data analytic platform for national-scale primary-care data. FLAIR contributes the missing combination for critical-care EHR-based CDS: modern longitudinal model classes, a clinically scoped data schema (CLIF) shared across sites, and a community-extensible task catalog under open governance. The five-attribute comparison is in Table 4.

6 Toward the Evidence the FDA Expects from EHR-Based CDS

Recent United States Food and Drug Administration guidance on clinical decision support software (**?**), together with the International Medical Device Regulators Forum's framework for the clinical evaluation of software as a medical device (**?**), asks developers of EHR-based CDS to make the basis of their recommendations transparent to the clinicians who use them and to demonstrate validation in the populations of intended use. Achieving that posture has historically been bottlenecked by the same access constraints described in the introduction: the relevant evidence is generated, if at all, one bilateral data use agreement at a time (Finlayson et al., 2021; Koch et al., 2024; Soin et al., 2025).

FLAIR's contribution to that posture is structural rather than declarative. Byte-identical task code stamped into every site's output makes the basis of a model's behavior re-inspectable rather than asserted; a reviewer can confirm that the cohort definition, exclusions, and outcome label that produced a site's metrics are the same ones that produced every other site's metrics, without taking that on trust. The per-site discrimination, calibration, and fairness JSONs surface the kind of population-level evidence that regulator-facing dossiers and clinician-facing documentation increasingly require, in a form that is mechanically auditable. And running across 12+ heterogeneous academic medical centers without data movement provides evidence of behavior under realistic distribution shift, where the literature has repeatedly shown that performance can silently degrade (Wong et al., 2021; Finlayson et al., 2021; Koch et al., 2024).

FLAIR is not a regulatory submission, not a substitute for prospective clinical trials, and not a guarantee of clinical utility. It is the layer of evidence between local model development and the regulator- or clinician-facing claims a developer eventually wants to make. That layer is currently missing for most published EHR CDS; making it routine is the goal.

7 Discussion

FLAIR combines three properties that, individually, have appeared elsewhere but together have not been operationalized at the scale ICU CDS development requires: federation across many heterogeneous sites without patient-data movement, a clinically scoped data schema (CLIF) that makes the analytic code portable, and an extension protocol that lets the task catalog grow by community pull request rather than by single-author fiat.

Saturation resistance through siloed data. A side effect of the model-to-data design is that the data underlying FLAIR's leaderboard never enters public hands. Test-set contamination of pretraining corpora, now a recurring problem for open benchmarks (Recht et al., 2019; Chekroud et al., 2024), is structurally limited, and the incentive for a single developer to overfit the leaderboard is weak because the data they would overfit to are not accessible to them. We view this not as a feature of FLAIR per se but as an emergent property of the access model.

Why evaluation, not training, federates safely. FLAIR federates evaluation rather than training, and the distinction is load-bearing for risk: aggregate metrics leak orders of magnitude less about patients than the weight updates exchanged in federated learning (Karargyris et al., 2023). The reciprocal risk, exposing a submitted model to sites, is bounded by central governance: the Benchmark Committee security-reviews the submitted code and model and validates it on synthetic

data before any site runs it, so model intellectual property and inversion risk stay contained. The asymmetry also bounds patient risk: only aggregate, cell-suppressed metrics leave a site and no model object ever does, so the federated outputs present far less re-identification surface than a shared model would — there is no per-site model for membership-inference or model-inversion attacks to target. Because those safeguards are fixed by design, participation can move from a per-pair data use agreement to a templated agreement, shrinking the legal latency that has historically capped multi-site validation at a handful of sites. Evaluation-only federation is also a deliberate stepping stone toward federated learning: the substrate it establishes — a common schema, a code-to-data Portal, and templated governance — is precisely what a future federated-training system would need, with each out-of-distribution consortium site then serving as a held-out validation node for federated-trained models.

Limitations. Adoption is bounded by the institutions that have implemented CLIF; non-CLIF sites cannot currently participate without an ETL effort. The current task suite is restricted to binary classification on structured EHR variables and excludes notes, waveforms, and images. Aggregate-only outputs constrain some fine-grained bias analyses; subgroups below the suppression threshold are not visible. FLAIR also federates evaluation only; orchestrating federated *learning* across sites is out of current scope.

Roadmap. Near-term work extends the suite to opt-in prospective silent-trial evaluation (Joshi et al., 2025), longitudinal calibration drift monitoring against site reference periods, and integration of additional CLIF tables (diagnoses, procedures) as the schema matures. Because CLIF standardizes not only the relational schema but also the permissible category vocabularies through its minimum common ICU data elements (mCIDE), a model defined once is reproducible across sites and over time: the same Portal submission and aggregation contract that runs on a retrospective extract applies unchanged to a live CLIF feed, so prospective, near-real-time benchmarking is a configuration of the existing protocol rather than a new system. The model-to-data primitive underpinning FLAIR is data-modality-agnostic; once CLIF stabilizes multimodal extensions, the same contract will apply to notes and waveforms. Federated learning is a feasible longer-term extension on the same foundation, but a materially heavier one: it requires live site-to-server connectivity, stronger security guarantees on the exchanged updates, and its own institutional review, rather than the one-directional aggregate-report exchange FLAIR uses today.

Conclusion. FLAIR makes privacy-preserving multi-site external validation of EHR-based clinical decision support a routine, reproducible operation. The library, task definitions, container, and report generators are open source (Apache 2.0, `github.com/clif-consortium/FLAIR`); researchers and clinicians are invited to add new benchmark tasks through the single-file PR template and two-site governance review described in §4, so the catalog grows with the field.

Required statements

Data and Code Availability. Source code, task definitions, Dockerfiles, and the report-generation library are released under Apache 2.0 at `github.com/clif-consortium/FLAIR`. MIMIC-CLIF is distributed via PhysioNet; consortium data are accessible only by running submitted code at each site through the FLAIR Portal. Baseline models are registered with and hosted on the FLAIR

Portal, which distributes them to sites; for open-access reproducibility the same weights and model cards are mirrored on the Hugging Face Hub and archived under a Zenodo DOI (**TODO:** DOI), and the CLIF-WorkBench container is published on GHCR and other public registries (**TODO:** tag).

IRB. Each consortium site obtained independent institutional review board approval; central protocol coordination is provided by the University of Chicago.

Funding. **TODO:** grants.

Author Contributions. CRediT taxonomy slots **TODO:** fill.

Conflicts of Interest. **TODO:** disclosures.

AI-Use Disclosure. **TODO:** name LLM(s) and the sections or artifacts they helped produce.

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A Supplementary appendix

S1. Count featurization specification

The FLAIR v1 baseline reduces each cohort window to a sparse count vector, following the count-featurization recipe of EHRSHOT and FEMR (Wornow et al., 2023; Steinberg et al., 2021) adapted to CLIF.

- **Token construction.** Every row of a CLIF event table (`vitals`, `labs`, `medication_admin_intermittent`, `medication_admin_continuous`, `respiratory_support`) is mapped to a deterministic `<table>.<category>` token. The dictionary is built by enumerating every schema-locked category present in MIMIC-CLIF; no per-task curation is applied.
- **Window.** For each cohort row, the featurizer counts every event in the same stitched encounter (`hospitalization_join_id`) recorded strictly before that row's `prediction_dttm`. This is a cumulative, point-in-time window — all prior history in the stay, from the encounter's first recorded event up to (but excluding) the prediction time — with no fixed look-back, matching the EHRSHOT/FEMR recipe. The exclusive `prediction_dttm` cutoff is the no-data-after boundary that prevents label leakage; note it is the *feature* cutoff, distinct from each task's prediction-grid start.
- **Representation.** Counts are accumulated into a `scipy` CSR sparse matrix (rows = cohort windows, columns = tokens); only non-zero counts are stored.
- **Missingness.** Absent token = zero count; there is no separate imputation step. A token unseen at training time is ignored at inference.
- **Variants not enabled.** FEMR's `CountFeaturizer` also supports ontology expansion, numeric-value deciles, string-value combinations, and time-binned counts; the FLAIR v1 baseline uses plain in-window counts only, so the baseline carries no external-ontology (Athena) dependency.

S2. Baseline hyperparameter grids

The XGBoost head is trained on MIMIC-CLIF using a 75/25 split on `hospitalization_join_id` (a stitched encounter never spans train and test), with model selection by validation AUROC over a grid search. Search spaces mirror the EHRSHOT count-baseline configuration.

- **XGBoost.** Gradient-boosted trees with `max_depth` $\in \{3, 6, \infty\}$, `learning_rate` $\in \{0.02, 0.1, 0.5\}$, and `num_leaves` $\in \{10, 25, 100\}$.

S3. Full baseline results on MIMIC-CLIF

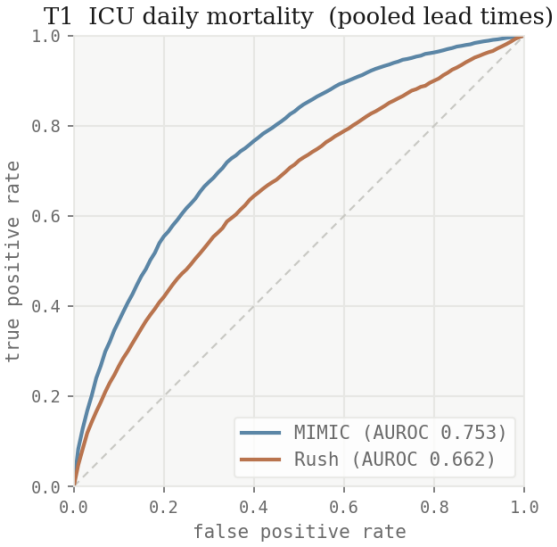
Per-task internal-test discrimination and calibration for the XGBoost baseline on the 75/25 stitched-encounter hold-out, scored in each task's report mode (§4). All metrics are mode-matched and stay-level; landmark rows are the lead-0 risk set. For the episodic and peak tasks (rare events trained with `scale_pos_weight`) the calibration slope is < 1 with a negative intercept, indicating over-confident scores: ranking is valid but probabilities need recalibration before use as decision thresholds. The landmark tasks are close to calibrated in slope. Peak (T4) omits calibration by design (the peak-trap). For context, the row-level (per-window) sepsis AUROC is 0.815, well above its mode-matched peak AUROC 0.693, the inflation that motivates mode-matched scoring.

Table 5: XGBoost count-featurization baseline: full mode-matched per-task metrics on the MIMIC-CLIF internal test split (75/25 hold-out on `hospitalization_join_id`). n is stays scored; prevalence is the mode-matched event rate; AUROC/AUPRC carry 1,000-iteration bootstrap 95% CIs.

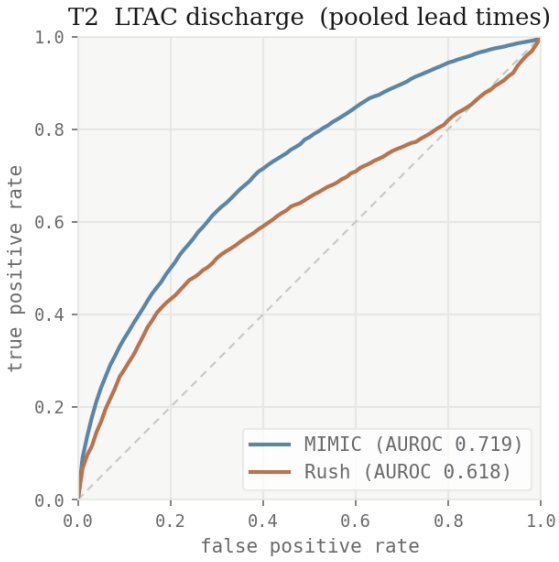
Task	Mode	n	Prev.	AUROC (95% CI)	AUPRC	Cal. slope	Cal. int.	Brier
T1, ICU daily mortality	landmark	18,195	0.134	0.864 (0.856–0.872)	0.587	1.19	−1.24	0.111
T2, LTAC discharge	landmark	18,195	0.065	0.789 (0.775–0.803)	0.288	1.01	−1.77	0.096
T3, Extubation failure 24 h	episodic	6,693	0.054	0.696 (0.665–0.719)	0.134	0.45	−1.86	0.070
T4, Sepsis 6 h	peak	19,943	0.063	0.693 (0.678–0.709)	0.134	—	—	—
T5, Unplanned ICU readmission	episodic	16,806	0.093	0.641 (0.626–0.658)	0.248	0.68	−1.89	0.151

S4. MIMIC-CLIF baseline curves

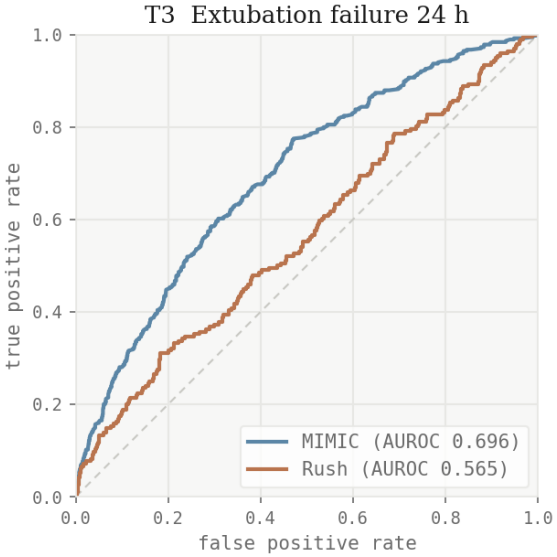
The per-task report PNGs the FLAIR library renders from the report JSONs (the same artifacts each consortium site emits), scored in each task's mode (§4). All plots are XGBoost. Figure 2 overlays both sites' ROC per task; the remaining panels (calibration, decision curves, landmark lead-time, fairness) are MIMIC-CLIF and follow each task's mode.



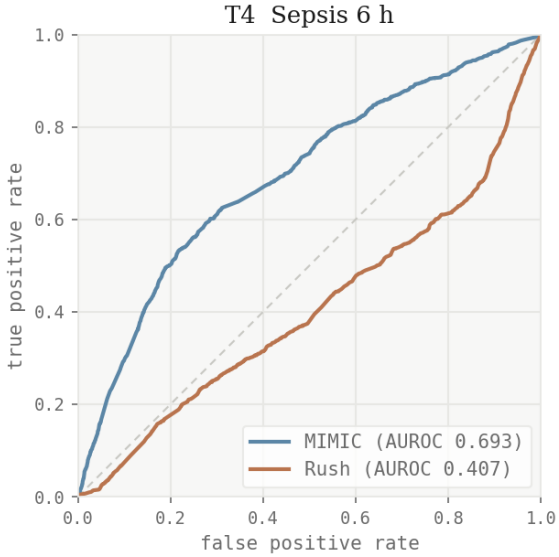
(a) T1 ICU daily mortality



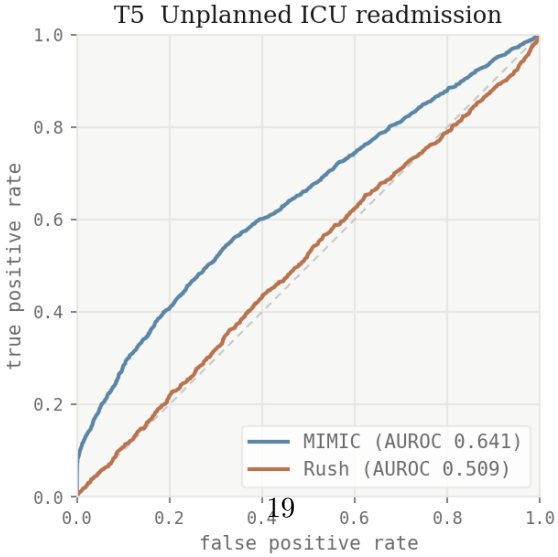
(b) T2 LTAC discharge



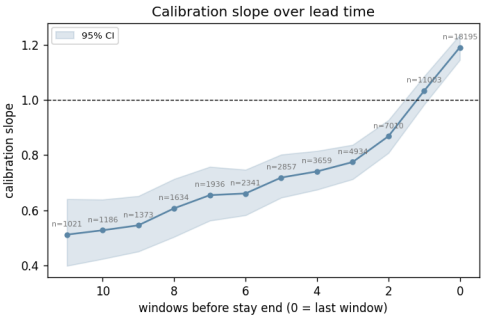
(c) T3 Extubation failure 24 h



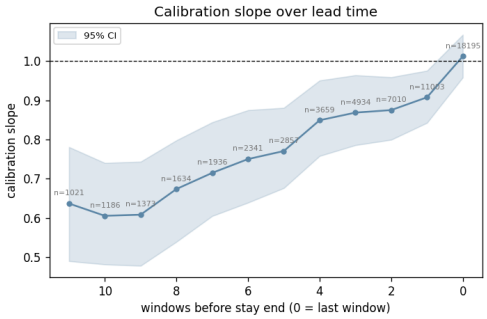
(d) T4 Sepsis 6 h



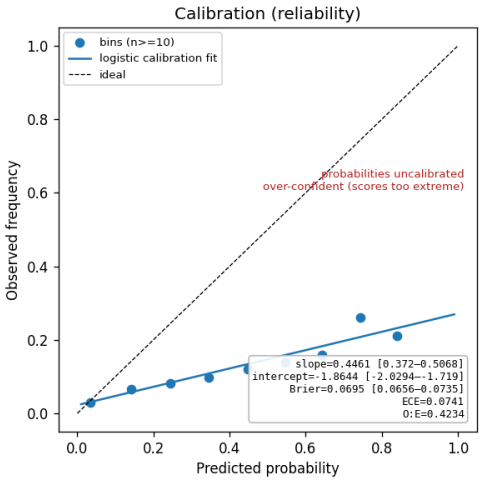
(e) T5 Unplanned ICU readmission



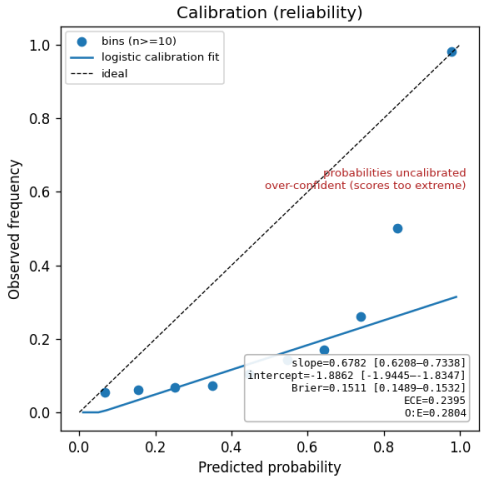
(a) T1 ICU daily mortality (landmark)



(b) T2 LTAC discharge (landmark)

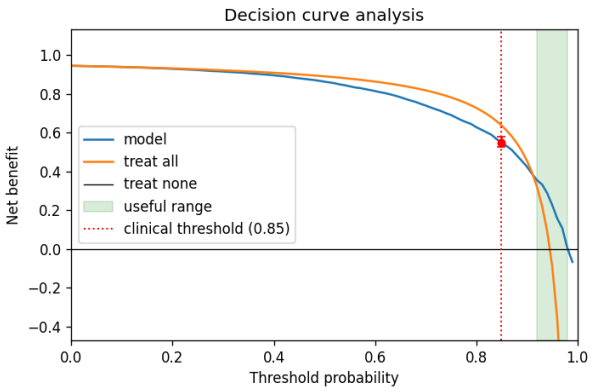


(c) T3 Extubation failure 24 h (episodic)

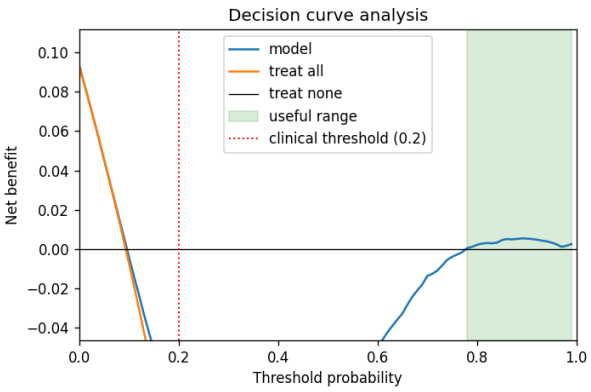


(d) T5 Unplanned ICU readmission (episodic)

Figure 3: **Reliability (calibration) curves.** Reliability points (bins with $n \geq 10$) against the diagonal, with the logistic recalibration slope/intercept in each title. The episodic tasks have slope < 1 (`scale_pos_weight` over-inflates predicted risk); the landmark tasks are closer to calibrated in slope. Peak (T4) omits calibration by design (the peak-trap).

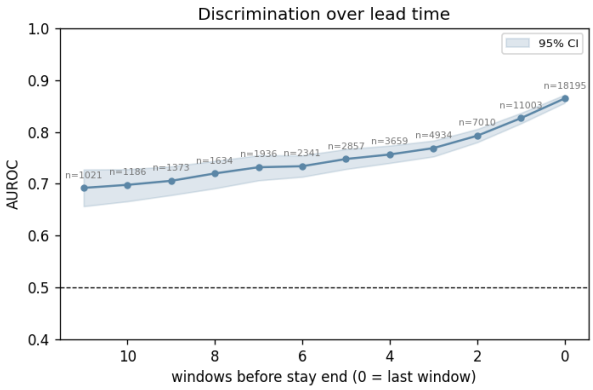


(a) T3 Extubation failure 24 h

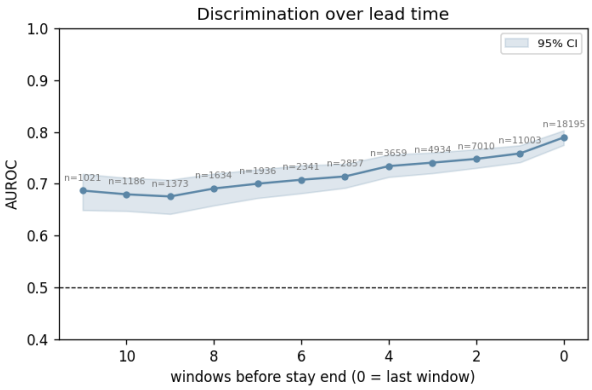


(b) T5 Unplanned ICU readmission

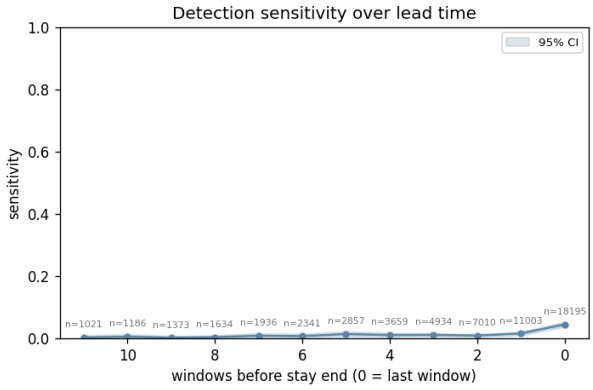
Figure 4: **Decision curve analysis (episodic tasks).** Net benefit of the model vs. treat-all / treat-none near the rare-event decision region; the red dotted line is the task's clinical threshold. The model beats both defaults across the useful threshold band.



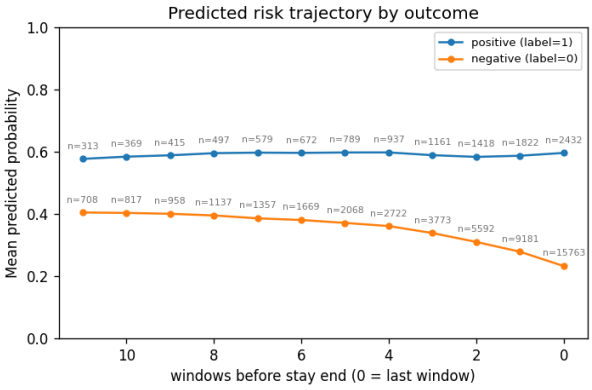
(a) T1 skill over lead time



(b) T2 skill over lead time

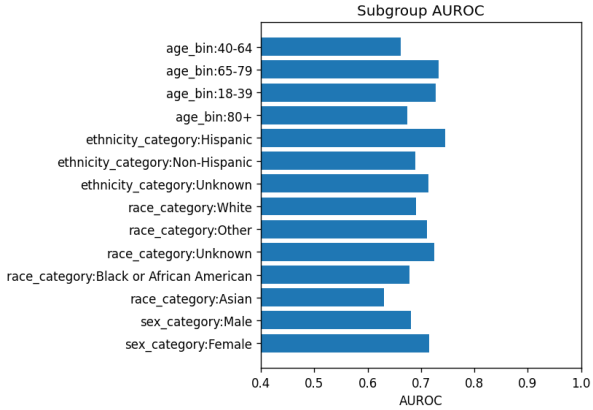


(c) T1 lead-time sensitivity

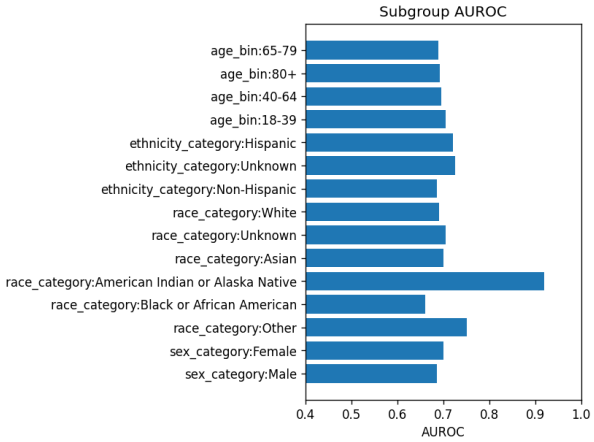


(d) T1 risk trajectory by outcome

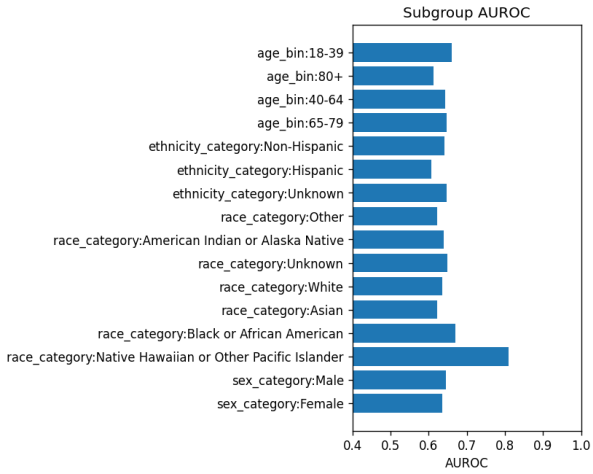
Figure 5: **Landmark lead-time views (T1, T2).** AUROC at each lead-time landmark (top), the share of event stays alarmed by lead time (bottom left), and the mean predicted-risk trajectory split by outcome (bottom right). These replace the single operating point for tasks scored as a sequence of daily risk sets.



(a) T3 Extubation failure 24 h



(b) T4 Sepsis 6 h



(c) T5 Unplanned ICU readmission

Figure 6: **Subgroup AUROC (fairness).** Per-subgroup AUROC across sex, race, ethnicity, and age bins with 95% CIs; subgroups with $n < 10$ are suppressed. Overlapping intervals across subgroups indicate no large discrimination-parity gap at the MIMIC-CLIF entry point.